Exact rho moments with unequal label probabilities

For every real $0 \le a \le 1$, Lean constructs a genuine PMF on all eight actual hash functions from Z/3 to two labels. Each vertex independently chooses the distinct nonzero increment $+1$ with probability a and $+2$ with probability $1-a$. Actual in-degree at vertex 0 has mean 1 and variance $1-(a^2+(1-a)^2)$. For probabilities $3/4$ and $1/4$, the variance is exactly $3/8$. [pcc:unverified]

Exact control	Mean	Variance
Uniform labels, 1/2 and 1/2	1	1/2
Unequal labels, 3/4 and 1/4	1	3/8

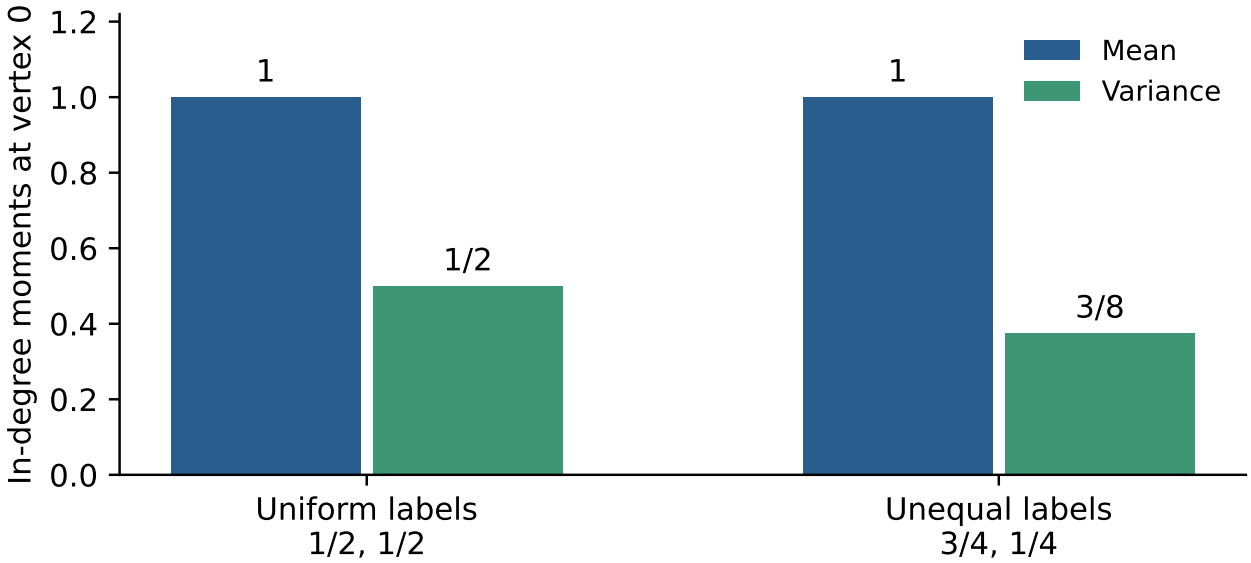

Figure 1: Recorded exact kernel controls for the complete eight-hash ensemble. No sampled data, timing or uncertainty interval.

The expanded whole walk project gate passes **43 declarations**, zero failures and zero trusted-native exceptions. Only propext, Classical.choice and Quot.sound are accepted. Lean 4.35.0 and Mathlib revision prefix `56a00585` are pinned. [pcc:unverified]

Construction and scope

The proof enumerates every actual hash through a proved bijection with three-bit tables. Kernel finite decision checks actual graph edges and in-degree counts. The product of independent vertex-label weights is proved nonnegative and normalized. PMF.ofFintype constructs the actual law, and ordinary real polynomial algebra establishes the moments for arbitrary a. The result is not fitted to observations. [pcc:unverified]

Source commit `a88c6fa0` precedes theorem packet `PCC-20261110-785828`. The combined binding is `weighted_variance_audit_certificate` in WalkWeightedAudit. Source/declaration hashes and all gate receipts are in `../../weighted-sources.json`; the local packet verdict and unsigned standing are recorded separately. Failed proof and registration attempts remain retained. [pcc:unverified]

This verifies the $1 - \sum p_i^2$ variance component in this exact ensemble. It does not derive the $1/\sqrt{1 - \sum p_i^2}$ collision-time penalty, a limiting constant, general support-size formulas, Hildebrand's averaged mixing theorem or practical hash-family behavior. No new native experiment or paid compute was run. Prior frozen source, protocol and data remain unchanged. [pcc:unverified]

Primary Galbraith chapter 14, version 2, section 14.2.5, pages 296–297, was re-read, including its ideal random-function assumption and separate conjectural collision estimates. The full PDF hash is in the source plan. A true weighted moment is not presented as a disproof of those heuristics. Scientific PCC standing remains unsigned until an independent signed verdict. [pcc:unverified]